

THE AI DISRUPTION

Predicting the future of the Tech Industry
using Machine Learning, and identifying
exactly which jobs will survive the decade.



Presented By: Abanoub Barayo

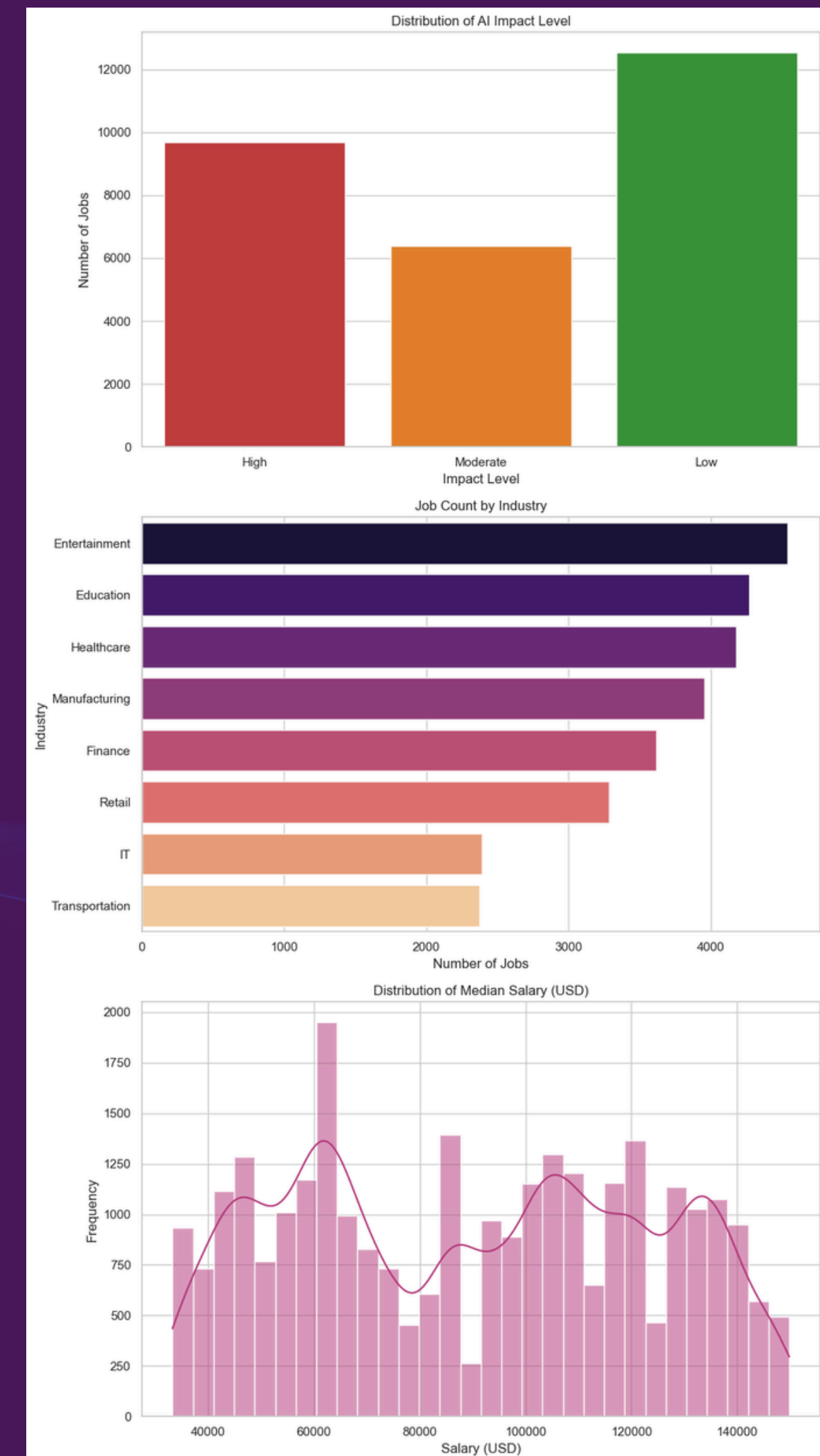
DECODING THE DATA

- **The Data:** Sourced from Kaggle. Thousands of tech industry jobs with 14 key features (Salary, Remote Work, Education, etc.).
- **The Goal:** Build a Machine Learning model capable of accurately predicting the AI_Impact_Level (High, Moderate, Low).



EXPLORATORY DATA ANALYSIS

- **The Big Picture:** Mapping initial trends across the Tech Industry to understand the baseline.
- **Risk Distribution:** Discovering that the majority of tech jobs currently fall into 'Low' or 'Moderate' AI impact categories.
- **Salary Insights:** Analyzing median salary distributions to identify financial incentives for corporate automation.
- **Hidden Patterns:** Uncovering early indicators that heavily digitized, remote-friendly roles face higher structural risks.



DATA LEAKAGE TRAP

- **The Problem:** Discovering 'synthetic' columns like Automation_Risk that act as an answer key.
- **The Danger of Leakage:** If left in, the model memorizes answers instead of learning true underlying patterns.
- **The Solution:** Dropping predictive shortcut columns to force the algorithm to rely strictly on raw structural data.
- **The Result:** A highly realistic, trustworthy Machine Learning model built on genuine tech industry dynamics.



MACHINE LEARNING JOURNEY

LOGISTIC REGRESSION

Failed to capture the complex, non-linear relationships in the tech industry data.

RANDOM FOREST

Established a strong baseline with **88%** accuracy

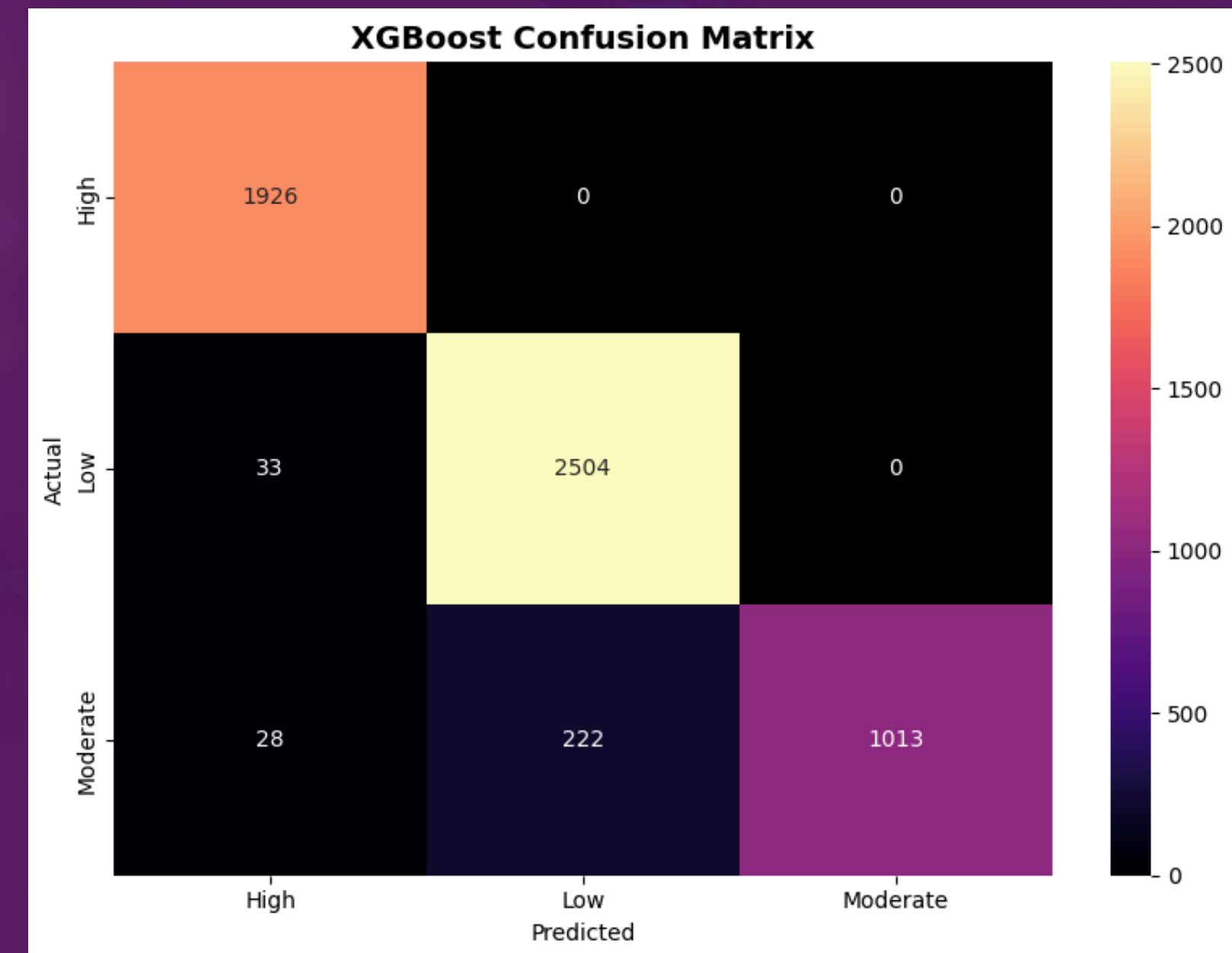
XGBOOST (THE CHAMPION)

Achieved an incredible **95%** accuracy after careful hyperparameter tuning to prevent overfitting.

Key Takeaway: The **XGBoost** model perfectly captured the structural realities of tech jobs without relying on data leakage.

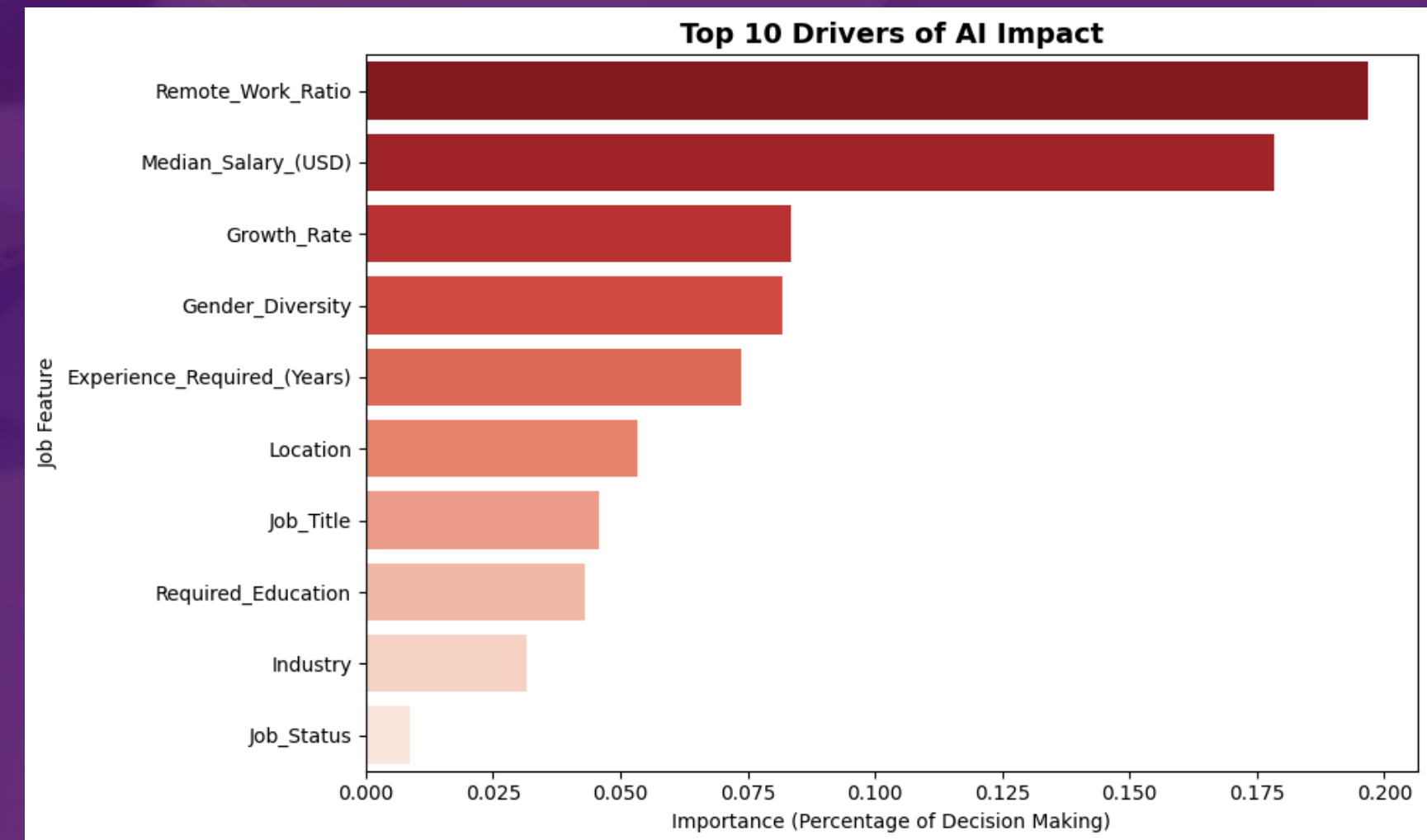
INSIDE THE BRAIN OF XGBOOST

- **Proving the Accuracy:** We know the model hit 95% accuracy, but a Confusion Matrix proves it isn't just making lucky guesses.
- **The Diagonal Line of Truth:** The heavy concentration of data along the center diagonal proves the model accurately mapped real-world job structures.
- **High Risk Precision:** When a job was genuinely 'High Risk', the model consistently caught it, meaning this algorithm can be trusted with new, unseen tech roles.



FEATURE IMPORTANCE: THE BIG REVEAL

- The Core Question: What makes a tech job vulnerable?
- The #1 Predictor: Remote Work Ratio dominates AI risk.
- The Structural Logic: 100% digital roles are the easiest to replicate.
- The Insight: AI replaces digital workflows, not just hard skills.



AI RISK MONITORING DASHBOARD

- The Goal: Translating complex Machine Learning insights into accessible business intelligence.
- The Audience: Designed for HR departments, executives, and non-technical stakeholders.
- The Solution: An intuitive 'Traffic Light' dashboard to monitor structural vulnerabilities at a glance.

DEMO

KEY INSIGHTS & DISCOVERIES



- **The Most Vulnerable Sector:** The IT Industry is overwhelmingly the most exposed to AI disruption, with over 70% of roles flagged as 'High Risk'.
- **The High-Risk Roles:** Jobs heavily reliant on digital creation and document analysis (e.g., Graphic Designers, Compliance/Legal) face the highest immediate threat.
- **The 'Safe' Zones:** Industries like Manufacturing and Retail show significantly lower AI vulnerability due to their reliance on physical-world interactions and logistics.

THANK YOU!



Github



AI Predictor

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